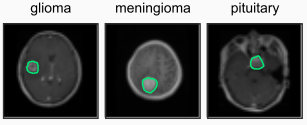
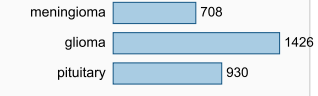
1 Data & pre-processing

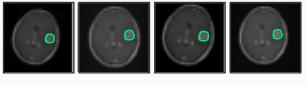


Figshare dataset [2,3]: 3,064 T1-CE slices, 233 patients; fields: image, label, patient ID, tumorMask (green)



- min–max intensity normalisation
- resize 512 → 256 (area); masks nearest
- grey → 3 channels, ImageNet statistics
- 224 × 224 network input

joint image–mask GPU augmentation



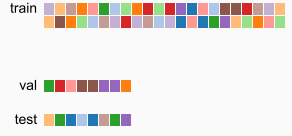
flip · rotate ±15° · scale ±10% · shift ±6% · contrast/brightness

G1 masks used

G4 class weights

2 Evaluation protocols

P1 image-level 80/10/10 (as in [21])



slices of the same patient (colour) fall in train and test → patient leakage; here all 160 test patients are also in train

P2 official patient-wise 5-fold CV



each patient in exactly one fold; ~10% of training patients form the validation set; 542–679 test slices per fold

P1 and P2 splits are fixed across all models and seeds

G2 leakage-free P2

3 Models & training

Reproduced baselines [21]

VGG16 / ResNet50

↓

one attention module
Coord · GCNet · SPNet · SRMNet
SSPANet · SE · CBAM

↓

GAP → FC

original recipe: Adam 1e-4, ≤30 epochs, early stopping, no augmentation

Proposed TG-SSPANet

backbone + TGSA × 3 (Fig. 3)

↓

mask-guided FPN decoder (Fig. 4)

↓

tumour-focused pooling gate

↓

GAP → FC

$$\mathcal{L} = \mathcal{L}_{cls} + \mathcal{L}_{seg} + 0.2\,\mathcal{L}_{align}$$

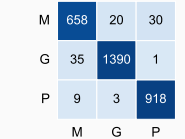
proposed recipe: AdamW + cosine, 40 epochs, augmentation, label smoothing, class weights, EMA, flip-TTA (also used for all ablations and same-recipe controls)

G3 multi-stage + ablation

G5 SE/CBAM added

4 Evaluation & explainability

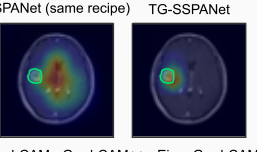
Classification



TG-SSPANet P2, 5 folds

Acc · F1 · κ · MCC · AUC · patient-level acc. patient-clustered bootstrap CIs corrected resampled t-tests · calibration

Explanations vs. tumour masks



Grad-CAM · Grad-CAM++ · EigenGrad-CAM pointing game · energy-in-mask · CAM-IoU

Faithfulness, ablation & cost

sham / decoy inpainting · context rings · location-only control · deletion/insertion · ablation · params · GMACs · latency

G1 quantitative XAI

G6 cost + variance

Research gaps of [21] addressed (red): G1 unused tumour masks · G2 patient leakage · G3 single-stage attention without ablation · G4 unhandled class imbalance · G5 SE/CBAM not evaluated · G6 unmeasured cost and run-to-run variance